BADR UNIVERSITY IN ASSIUT
Faculty of Artificial Intelligence & Data Management

Machine Learning Capstone Project

Regression | Classification | Clustering

California Housing Dataset • Project Report

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Google Colab Notebook:

https://colab.research.google.com/drive/1pXIdQXmVC2CWclSmOsrkLcm6-CwGs_iM?usp=sharing

I Introduction

Project Goal

This project walks through the full machine learning pipeline on a single real-world dataset: data acquisition, cleaning, exploratory analysis, preprocessing, modeling with three different algorithm families, evaluation, and reporting. Using the **California Housing** dataset, three tasks are solved:

- **Regression** – predict the median house value of a district with **Linear Regression**.
- **Classification** – label a district as *high-value* or *low-value* with **Logistic Regression**.
- **Clustering** – discover natural groups of districts with **K-Means**, without using any label.

Data Description

The data comes from the 1990 U.S. Census and is available on Kaggle (*California Housing Prices*). Each row describes one census block group (district) in California. The file `housing.csv` contains **20,640 rows** and **10 columns**.

20,640	10	13	207
Districts (rows)	Original columns	Features after encoding	Missing values

Feature	Type	Description
longitude / latitude	Numeric	Geographic location of the district
housing_median_age	Numeric	Median age of houses in the district
total_rooms / total_bedrooms	Numeric	Total rooms / bedrooms in the district
population / households	Numeric	Residents / households in the district
median_income	Numeric	Median household income (tens of thousands of $)
ocean_proximity	Categorical	<1H OCEAN, INLAND, NEAR OCEAN, NEAR BAY, ISLAND
median_house_value	**Target**	Median house value in $ (regression target)

II Data Preprocessing & EDA

Cleaning Steps

Step	Action	Reason
Missing values	`total_bedrooms` had 207 NaNs, filled with the **median**	Median is robust to the skewed distribution
Duplicates	0 duplicate rows found	No action needed
Encoding	One-Hot Encoding of `ocean_proximity` (5 new columns)	Nominal feature with no order
Scaling	`StandardScaler` (fit on train only for supervised tasks)	Required for K-Means; avoids data leakage
Split	80% train (16,512) / 20% test (4,128), `random_state=42`	Fair, reproducible evaluation

III Modeling and Results

1. Regression – Linear Regression

A **Linear Regression** model was trained on the scaled training set to predict `median_house_value` and evaluated on the 4,128 test districts.

4.91×10^9	$70,061	0.625
MSE	RMSE	R^2 Score

Interpretation. An R^2 of **0.625** means the model explains about **62.5%** of the variance in house values. The RMSE of about **$70,061** is the typical prediction error, roughly 34% of the average house value ($206,856). The model captures the overall trend but under-predicts the most expensive districts, which is expected because the relationship is not perfectly linear and values are capped at $500,001. Ridge ($R^2$ = 0.6255) and Lasso (R^2 = 0.6254) were also tried as extras and gave practically identical results,

2. Classification – Logistic Regression

To create a binary target, a new label `high_value` was defined: **1** if the district's house value is above the dataset median ($179,700), otherwise **0**. This gives two perfectly balanced classes. A stratified 80/20 split was used.

83.79%	0.83	0.84	0.84
Accuracy	Precision (high)	Recall (high)	F1-score

Confusion Matrix Results

The model correctly classified 3,459 out of 4,128 test samples, achieving an accuracy of 83.79%. It misclassified 669 samples, including 347 low-value samples predicted as high- value and 322 high-value samples predicted as low-value.

3. Clustering – K-Means

All 13 features (the target and the `high_value` label were excluded) were standardized and K-Means was run for K = 2 to 10. The **Elbow Method** plots inertia against K.

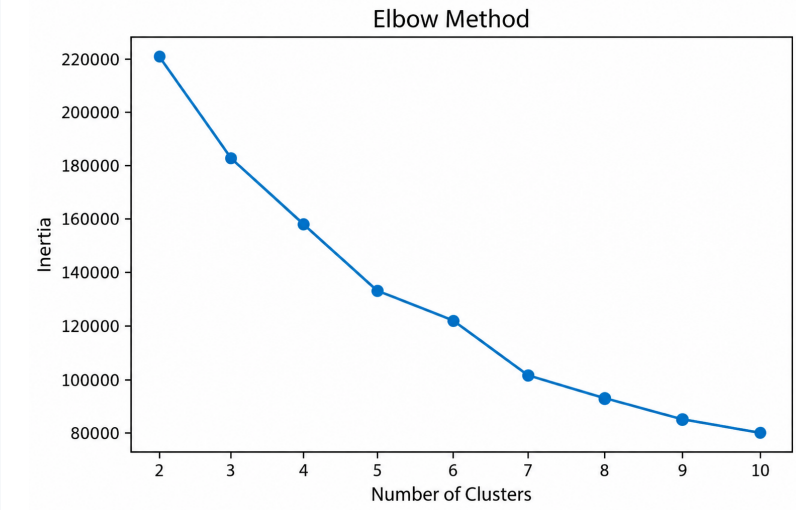


Figure 5 – Elbow Method: inertia vs number of clusters

Choice of K. Inertia drops steeply from K = 2 to K = 5 and the slope becomes noticeably flatter after K = 5 and again after K = 7. The curve has no sharp elbow, so **K = 5** was chosen as a balance between compactness and interpretability (more clusters would be harder to explain with little extra gain).

K = 5	0.361	132,965
Chosen clusters	Silhouette Score	Final inertia

Silhouette Score = 0.361. This is a moderate value (the range is –1 to 1): clusters are reasonably separated but overlap, which is natural for geographic and economic data that changes gradually from place to place.

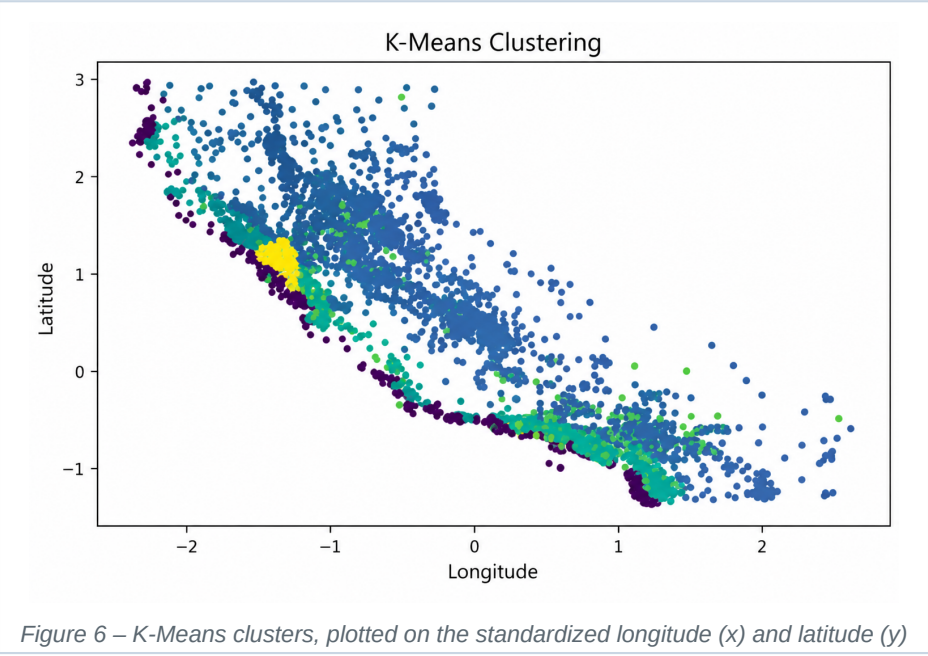


Figure 6 – K-Means clusters, plotted on the standardized longitude (x) and latitude (y)

Cluster Interpretation

Cluster	Size	Avg. income	Avg. value	Main characteristic
0	2,579	4.0	$248,441	Coastal districts – almost all NEAR OCEAN
1	6,048	3.2	$122,823	Affordable INLAND districts, lowest income
2	8,500	4.2	$239,263	Largest group – <1H OCEAN suburban districts
3	1,296	4.2	$214,123	Very dense districts (avg. population 4,322), newest housing (15.5 yrs)
4	2,217	4.2	$259,225	San Francisco Bay Area (NEAR BAY), oldest housing (38 yrs), most expensive

The clusters are driven mostly by **location and proximity to the ocean**, together with income and housing characteristics. Cluster 1 (inland, low income) is clearly separated from the expensive coastal clusters 0 and 4, while cluster 3 isolates unusually large, dense districts. Note that the plot uses only two of the 13 features, so some overlap between colors is expected.

IV Conclusion

Best-Performing Model for Each Task

Task	Model	Main result
Regression	Linear Regression	R^2 = 0.625, RMSE = $70,061
Classification	Logistic Regression	Accuracy = 83.79% (balanced precision/recall)
Clustering	K-Means (K = 5)	Silhouette = 0.361, five interpretable regional groups

Challenges Encountered

- Missing values in `total_bedrooms` and the categorical `ocean_proximity` column required cleaning and encoding.
- The target is capped at $500,001, which limits what a linear model can learn for expensive districts.
- The Elbow curve is smooth with no clear elbow, so K had to be chosen with judgment.
- The dataset has no natural binary label, so a `high_value` label was engineered from the median.

Potential Next Steps

- Try non-linear models such as **Random Forest** or **Gradient Boosting** (XGBoost / LightGBM) to improve R^2.
- Add engineered features such as rooms per household, bedrooms per room and population per household.
- Apply **PCA** before clustering and compare K choices with Silhouette analysis.
- Use cross-validation and hyper-parameter tuning for more reliable estimates.

Project Links

Resource	Link
Google Colab Notebook	https://colab.research.google.com/drive/1pXIdQXmVC2CWclSmOsrkLcm6-CwGs_iM?usp=sharing
GitHub Repository	https://github.com/Aboleila-Engineer/ML-Capstone-Project
Dataset	Kaggle – California Housing Prices (1990 U.S. Census)